Continual Concept Erasure and Restoration of Diffusion Models

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Abstract

As text-to-image generative models are deployed at scale, ensuring they cannot produce restricted, harmful, or legally protected content has become a critical safety requirement. Most concept-erasing techniques implicitly assume a *static* set of forbidden concepts. In real-world deployments, however, restrictions evolve due to changing policies, licensing agreements, and safety requirements, making *continual erasing* a practical necessity. While a few recent works have explored sequential concept removal, no standardized protocol exists for comparing continual erasure methods under consistent concept sequences, prompt sets, and evaluation metrics, leaving it unclear which methods degrade gracefully over repeated erasures and which do not. Moreover, *restoration*, the selective re-enabling of a previously erased concept without undoing other erasures, has never been formally defined or evaluated, despite being essential for long-term model governance. To close both gaps, we (i) introduce *CORE-Bench*, a standardized benchmark for jointly evaluating continual concept erasure and restoration across multiple domains; (ii) formally define the restoration task and propose a *general constrained counter-edit formulation* that instantiates selective restoration for both training-based and closed-form erasers; and (iii) demonstrate, through systematic benchmarking, the *clear advantage of attention-based closed-form editors for selective restoration with minimal leakage* over training-based alternatives on open-source diffusion models.

1 Introduction

Text-to-image generative models [28, 29] have transformed creative and commercial applications, yet they carry significant safety, ethical and compliance risks. Since these models are trained on large datasets scraped from the internet, which frequently contain copyrighted artwork, harmful imagery and sensitive content, they may unintentionally generate restricted material. This has motivated a growing body of research on *concept erasing* [4, 5, 9, 17, 21, 22], which aims to remove specific concepts from a model while preserving

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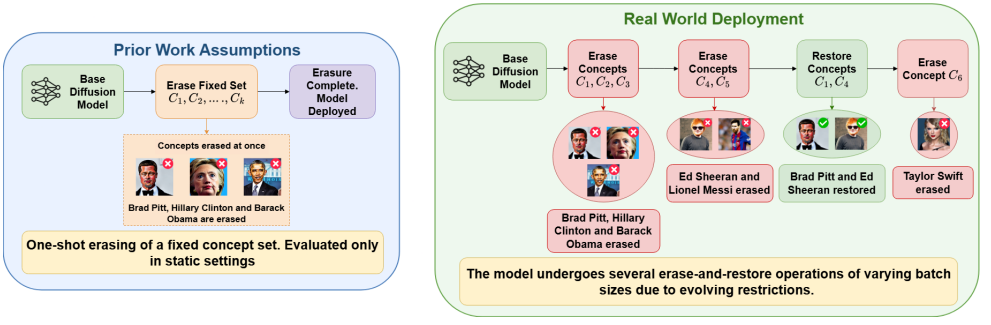


Figure 1: The gap between prior erasure works and deployment scenarios. Prior works assume erasure requests arrive together (e.g, erase Brad Pitt, Hillary Clinton and Barack Obama) while in real world scenarios, the requests can arrive over a long period of time with varied batch sizes (e.g, first erase Brad Pitt (C_1), Hillary Clinton (C_2) and Barack Obama (C_3). Next, erase Ed Sheeran (C_4) and Lionel Messi (C_5)). Additionally, a restricted concept set (e.g, Brad Pitt (C_1) and Ed Sheeran (C_4)) might be restored between erasures at a later time thus making restoration a critical problem.

its general utility. While effective at suppressing targets in controlled settings, these methods overwhelmingly assume that the set of forbidden concepts is *static and known in advance*, an assumption that rarely holds in practice.

In real-world deployments, safety and legal requirements evolve [3, 33], new content may become restricted or court rulings may expand copyright protections. A practical system, such as the one shown in Fig. 1, must therefore support *continual* concept erasing, where edits are applied sequentially over time. This introduces challenges absent from the one-shot setting such as interference between successive edits and cumulative semantic drift. Only recently has this realistic scenario received attention [9], but the evaluation scope is still limited to a very few sequentially erased concepts [9]. Prior benchmarks [1, 26, 37] assess erasure strength but exclusively in single-shot settings, where a fixed concept set is removed in one stage. [18] studies utility degradation under continual unlearning and proposes regularization to mitigate parameter drift, but is restricted to training-based methods [4, 16, 19] and does not cover closed-form or parameter-efficient approaches. *To the best of our knowledge, there is currently no standardized protocol that evaluates continual erasure.*

A second, largely overlooked challenge is *restoration*; selectively re-enabling a previously erased concept without undoing other removals. This arises naturally when policies or licensing agreements [3, 33] evolve and a concept restricted today may become permissible tomorrow. Restoration has been explored in adjacent areas. [25] alternates forgetting and restoration to improve generalization in deep neural networks, and [27] incorporates restoration into an unlearning evaluation framework for large language models. In the image generation domain, however, restoration remains formally undefined.

Our main contributions are:

1. We introduce *CORE-Bench*, a benchmark for continual concept erasing, covering three domains and repeated erasure sequences upto twenty-five concepts, under a unified eval-

uation protocol.
2. We formalize selective concept restoration in diffusion models as a constrained counter-edit problem, with concrete instantiations for training-based and closed-form erasers.
3. We introduce a restoration evaluation protocol with three scenarios; single-concept, multi-concept and sequential restore-erase.
4. We provide benchmarking using several baselines spanning three concept-erasure paradigms.

2 Related Works

In this section, we briefly discuss concept erasure methods which we group into three classes based on their erasure strategy: (i) training based methods, (ii) closed-form editing, and (iii) parameter-efficient fine-tuning (PEFT).

Training based methods employ iterative gradient-based optimization to modify model weights. ESD [4] performs post-hoc model editing by fine-tuning the model. The ESD-x variant edits the cross-attention weights [34] while ESD-u variant edits non-cross-attentional parameters. For our work, we use ESD-x. Concept Ablation [16] prevents the generation of the erased concept by minimizing the KL divergence with the anchor concept distribution, while CRCE [38] and CoGFD [23] leverage LLMs to generate the retain and anchor sets to be used in the noise prediction loss. CCRT [9] is the first work to study sequential concept removal in diffusion models. It performs continual erasing via iterative fine-tuning, using a knowledge distillation objective between a frozen teacher (base) and a trainable student model along with a preservation set. However its capability is limited only upto three sequential erases. ACE [36] introduces erasure guidance into conditional and unconditional noise predictions to prevent generation and reintroduction of erased concept via image editing. FADE [32] performs fine grained erasure via adjacency-aware unlearning.

Closed-Form Editing methods compute erased weights in a single closed-form step, without gradient-based training. UCE [6] and TIME [24] perform erasure via a closed-form modification of the key and value matrices of the cross-attention layer [34]. MACE [21] extends closed-form attention editing with concept-specific LoRA [14] fine-tuning. Instead of naively summing the adapters, MACE fuses multiple adapters by solving a closed-form optimization that preserves individual erasures. SPEED [20] extends closed form editing by performing edits in the null-space of retain set while RECE [8] performs adversarial fine-tuning to improve UCE [6].

Parameter Efficient Fine-Tuning (PEFT) methods circumvent the need for full-model updates by introducing lightweight adapters [14]. These adapters suppress target concepts while keeping the base model weights frozen. SPM [22] adopts a parameter-efficient fine-tuning strategy in which each concept is erased by training a one-dimensional LoRA [14] while keeping the base model frozen. Adapters compose linearly, enabling efficient accumulation of edits without retraining. GLoCE [17] performs localized concept erasure by introducing lightweight low-rank matrices to the output layers of the diffusion model [28]. To selectively erase concepts, it creates a gating mechanism whose parameters are determined by a few-shot inference only approach. Receler [15] utilizes adversarial prompt learning and concept-localized regularization to learn adapters that are robust against paraphrased prompts and preserve non-target concepts.

3 Continual Concept Erasure

Rather than evaluating only methods that explicitly target continual erasure, we study the full spectrum of erasure paradigms; training-based, closed-form and PEFT, under a unified continual protocol, since understanding their failure modes under repeated edits is a prerequisite for building better systems. We therefore adapt each method to the continual setting using the natural extension of its own update rule, described below, and evaluate training-based methods, ESD-x [9] and CCRT [9], closed-form editing techniques, UCE [5] and MACE [21], and PEFT-based approaches, SPM [22] and GLoCE [17].

We denote by θ_i the model obtained after the i^{th} continual erasure update, where θ_0 is the original pretrained diffusion model. Let x_t denote the noisy latent at diffusion timestep t, and let $\varepsilon_\theta(\cdot)$ denote the noise predictor parameterized by θ. Given θ_i with i erased concepts, the *continual erasure* problem seeks θ_j $(j > i)$ that removes $j - i$ additional concepts while (i) preserving other concepts and (ii) preventing reintroduction of previously erased concepts, a constraint absent from all static formulations.

Training-based methods: Training-based methods iteratively fine-tune model weights via gradient-based optimization. Under the continual protocol, each method reuses the most recently edited checkpoint θ_{i-1} as the starting point for the next erasure, rather than returning to θ_0. But this natural extension introduces the risk of interfering with prior edits, causing previously erased concepts to resurface (concept resurgence [2, 31]).

For ESD-x [9], the loss equation $\mathcal{L}_{\text{ESD}}$ for concept c_i at i^{th} continual erasure step is:

$$\mathcal{L}_{\text{ESD}} = \left\|\varepsilon^i_\theta(x_t, c_i) - (\varepsilon^{i-1}_\theta(x_t, \varnothing) - \lambda\left(\varepsilon^{i-1}_\theta(x_t, c_i) - \varepsilon^{i-1}_\theta(x_t, \varnothing)\right))\right\|_2^2 \tag{1}$$

$\varnothing$ is the unconditional prompt embedding and λ controls erasure strength. Since ESD-x operates directly in noise-prediction space, repeated applications can accumulate drift across the full parameter space, making it susceptible to inter-edit interference under repeated erasures.

CCRT [9] mitigates this by augmenting the ESD-x objective $\mathcal{L}_{\text{ESD}}$ with a calibration term that anchors the model to θ_0 on a set of preserved prompts $\mathcal{P}_{\text{ret}}$:

$$\mathcal{L}_{\text{CCRT}} = \mathcal{L}_{\text{ESD}} + \lambda\, \mathbb{E}_{p\in\mathcal{P}_{\text{ret}}} \left\|\varepsilon_\theta(p) - \varepsilon_{\theta_0}(p)\right\|_2^2 \tag{2}$$

While this reduces semantic drift on retained concepts, CORE-Bench evaluates whether this calibration remains effective under repeated erasures.

Closed-form editing methods: For closed-form methods, under the continual protocol, each update is initialized from the previous edited checkpoint θ_{i-1}, so edits accumulate compositionally.

For UCE [5], the i^{th} update solves:

$$\min_{\theta_i} \lambda_1 \|\theta_i X^i_{\text{tar}} - \theta_{i-1} X^i_{\text{map}}\|_2^2 + \lambda_2 \sum_{X\in\mathcal{C}_{ret}} \|\theta_i X - \theta_{i-1} X\|_2^2 \tag{3}$$

where X^i_{tar} and X^i_{map} are target and mapping text embeddings and $\mathcal{C}_{ret}$ is the retained concept set. The retention term anchors unrelated directions but does not explicitly constrain previously erased concepts, leaving them potentially vulnerable under repeated erasures.

For MACE [21], concept-specific LoRA updates $\Delta\theta_i$ are computed on θ_{i-1} and fused:

$$\theta_i = \arg\min_\theta \left\|\theta X_i - (\theta_{i-1} + \Delta\theta_i) X_i\right\|_2^2 + \lambda \left\|\theta X_{\text{ret}} - \theta_{i-1} X_{\text{ret}}\right\|_2^2, \tag{4}$$

where X_i are embeddings of target concept c_i and X_{ret} are prior-preserving embeddings. MACE's fusion step explicitly protects preserved directions, which may confer better stability under continual editing than UCE.

Parameter-efficient fine-tuning (PEFT) methods: PEFT methods introduce lightweight, concept-specific adapters while keeping base model weights frozen. This architectural choice has a direct implication for the continual setting. Since edits are *modular and non-destructive*, PEFT methods are structurally immune to the interference and drift that affect weight-editing approaches. CORE-Bench tests whether this theoretical advantage translates to empirical stability at scale.
SPM [22] attaches a one-dimensional LoRA adapter $(v^i_{\text{reg}}, v^i_{\text{sig}})$ per target concept c_i in C_{tar}:

$$y = \theta_0 x + \sum_{i \in C_{tar}} \left(\gamma^i (v^i_{\text{reg}})^\top x\right) v^i_{\text{sig}}, \tag{5}$$

γ^i is a concept-specific gating scalar. x represents the input activation to the hidden layer, which is being updated to y. Continual erasure reduces to appending a new adapter to the sum, with no retraining of existing adapters.
GLoCE [17] injects gated low-rank adaptations (P, b) into the model output layers:

$$f(z) = \big(1 - s(z)\big)z + s(z)(Pz + b), \tag{6}$$

where $s(z)$ is a concept-dependent gating function and z is the output of the hidden layer, learned via few-shot inference, that activates only for target concept embeddings. Since the gate is localized to the target concept's representation, GLoCE is designed to minimize collateral activation on unrelated concepts.

4 Selective Concept Restoration

While continual erasure has begun to receive attention, the complementary problem of *selective concept restoration* remains almost entirely unexplored in the context of generative models. In practice, the same regulatory and licensing frameworks that mandate erasure also create conditions under which restrictions are lifted. Consider, for instance, a scenario where a character's copyright expires or a previously restricted artist grants new licensing rights. A deployed model should be updated to reflect this change without requiring full retraining or undoing unrelated restrictions.

Problem definition and Challenges: Given an erased model θ_i with erased set $\mathcal{C}_{\text{tar}} = \{c_1, \ldots, c_i\}$, we define *selective restoration* as finding a restored model $\theta_i^{c_r}$ that re-enables a target concept $c_r \in \mathcal{C}_{\text{tar}}$ such that: (i) c_r regains its original generative behavior as in base model θ_0, (ii) all other erased concepts $\mathcal{C}_{\text{tar}} \setminus \{c_r\}$ remain suppressed, and (iii) the model's behavior on unrelated retained concepts is preserved.
A naive solution is to revert to an earlier checkpoint θ_{r-1}, before concept c_r was erased. This trivially restores c_r but simultaneously undoes *all* erasures that followed, violating the restrictions that remain legally or ethically necessary. The restoration problem thus requires the *selective* reintroduction of one concept while holding all other edits fixed, which makes it challenging. The goal is to enable *authorized, operator-controlled* updates of the model, rather than attacking it to bypass earlier restrictions.

Restoration under PEFT methods. For PEFT methods (SPM, GLoCE), restoration reduces to removing the concept-specific adapter for c_r from the active adapter set, a straightforward operation given the modular, non-destructive nature of adapter-based editing. While simple and effective, this approach has an *important deployment limitation*. It requires that model weights and adapter parameters remain inaccessible to end users. In open-weight settings, the same modularity that enables easy restoration becomes a vulnerability, a malicious user who can access the adapter files can disable any adapter to bypass erasure without authorization. PEFT-based restoration is therefore best suited to API-gated or closed deployment environments.

Restoration under weight-editing methods. Training-based and closed-form editing methods are *well suited for open-weight* deployment, as they perform edits directly to the model weights. *However, edits corresponding to multiple concepts can become entangled over time, making selective restoration non-trivial*. Moreover, restoration procedures are inherently method-specific since different erasure methods modify parameters in various ways, leading to different forms of entanglement. As a result, a single implementation is unlikely to apply unchanged across different erasure approaches. We address this by proposing a *general principle* of restoration as a constrained counter-edit and instantiating it for ESD-x which is training-based and two closed-form editors, UCE and SPEED.

Training-based methods: Let θ_0 denote the base model [28] and θ_i the model after sequentially erasing $\{c_1,\ldots,c_i\}$. Let $\mathcal{R}(c_r)$ be a prompt set specific to the restored concept c_r, $\mathcal{P}_{tar}$ the set of prompts for all erased concepts, and $\mathcal{P}_{ret}$ prompts for retained concepts. We minimize:

$$\begin{aligned}\mathcal{L}_{\text{restore}} = \mathbb{E}_{p\in\mathcal{R}(c_r)}\left\|\varepsilon_\theta(x_t,p) - \underline{\varepsilon_{\theta_0}(x_t,p)}\right\|_2^2 \\ + \lambda_1\mathbb{E}_{p\in\mathcal{P}_{\text{tar}}\setminus\mathcal{R}(c_r)}\left\|\varepsilon_\theta(x_t,p) - \underline{\varepsilon_{\theta_i}(x_t,p)}\right\|_2^2 \\ + \lambda_2\mathbb{E}_{p\in\mathcal{P}_{\text{ret}}}\left\|\varepsilon_\theta(x_t,p) - \underline{\varepsilon_{\theta_i}(x_t,p)}\right\|_2^2.\end{aligned} \tag{7}$$

The underlined terms mark the restoration-specific changes to the original ESD-x erasure objective. For prompts of the restored concept, $p\in\mathcal{R}(c_r)$, the edited model is pulled back toward the base-model prediction $\varepsilon_{\theta_0}(x_t,p)$, thereby reversing the erasure only for c_r. For all non-restored erased concepts and retained concepts, the model is instead constrained to match the deployed erased checkpoint $\varepsilon_{\theta_i}(x_t,p)$. Thus, ESD-x restoration selectively aligns the restored concept with the base model while anchoring other concepts to the erased model.

Closed-form editing: For UCE [5], restoration can be implemented by reusing the same closed-form update rule used for editing cross-attention weights [34], but with modified targets. Starting from the deployed erased weights θ_i, we solve for restored weights θ that (i) match the base-model projection of restored concepts and (ii) match the erased-model projection of all concepts that should remain unchanged:

$$\begin{aligned}\min_\theta \quad \alpha_r\sum_{c\in\mathcal{C}_r}\left\|\theta c - \underline{\theta_0 c}\right\|_2^2 + \alpha_e\sum_{c\in\mathcal{P}_{\text{tar}}\setminus\mathcal{C}_r}\left\|\theta c - \underline{\theta_i c}\right\|_2^2 \\ + \alpha_k\sum_{c\in\mathcal{P}_{ret}}\left\|\theta c - \underline{\theta_i c}\right\|_2^2 + \lambda\left\|\theta-\theta_i\right\|_2^2.\end{aligned} \tag{8}$$

The underlined terms mark the restoration-specific modification to the original UCE erasure update. For restored concepts, $c\in\mathcal{C}_r$, the edited projection is aligned with the base-model response, thereby reversing the erasure only for the selected concepts. For non-restored erased concepts and retained concepts the edited projection is instead constrained to match

the deployed erased model $\theta_i c$. Since UCE operates directly on the cross-attention key and value matrices, the restoration update is computed in a single closed-form step, with no gradient-based training.

We also instantiate the same principle for SPEED [20], which performs closed-form attention editing through a null-space preservation mechanism. Let θ_i denote a SPEED-edited cross-attention weight matrix after continual erasure, and let θ_0 denote the corresponding base-model weight. For a restored concept set $\mathcal{C}_r$, SPEED restoration computes a counter-edit that moves restored concept responses toward the base model while restricting the update to directions that minimally affect preserved concepts. If P denotes the null-space projector estimated from the non-restored erased concepts and retained concepts, we write:

$$\begin{aligned} \min_{\Delta} \quad & \sum_{c \in \mathcal{C}_r} \left\| (\theta_i + \Delta P)c - \underline{W_0 c} \right\|_2^2 + \lambda \|\Delta P\|_2^2 \\ \text{s.t.} \quad & (\Delta P)c \approx 0, \qquad \forall c \in \underline{(\mathcal{P}_{\text{tar}} \setminus \mathcal{C}_r) \cup \mathcal{P}_{\text{ret}}}. \end{aligned} \tag{9}$$

The underlined terms indicate the restoration-specific modification to the original SPEED update. Instead of mapping the selected concepts toward an anchor or null response, restoration maps concepts in $\mathcal{C}_r$ back toward the base-model response $\theta_0 c$. At the same time, the preservation set is expanded to include all non-restored erased concepts, $\mathcal{P}_{\text{tar}} \setminus \mathcal{C}_r$, together with the retained concepts $\mathcal{P}_{\text{ret}}$. The null-space projector P therefore restricts the counter-edit to directions that minimally affect concepts that should remain unchanged.

Eqs. (7), (8), and (9) instantiate the same principle under different editing paradigms. Together, they demonstrate that selective restoration is tractable under both training-based and closed-form methods. Our experiments reveal important differences in how well each method satisfies condition (ii) in the problem formulation. UCE and SPEED maintain low leakage on other erased concepts after restoration, whereas ESD-x suffers significant concept resurgence [6, 7, 31].

5 CORE-Bench: Continual Concept Restoration & Erasure Benchmark

We introduce **CORE-Bench** (**C**ontinual c**O**ncept **R**estoration and **E**rasure Benchmark), a standardized benchmark designed to evaluate concept erasing and restoration methods under realistic long-horizon, sequential editing conditions. CORE-Bench tests erasure methods in two ways : (i) how effectively a method erases new targets without degrading previously erased concepts or unrelated retained content as the edit sequence grows, and (ii) how selectively a method can restore a previously erased concept without reactivating other forbidden concepts or destabilizing retained ones.

Evaluation data: All experiments follow a unified configuration specifying the concept domain, erasing regime, continual horizon, prompt templates and evaluation seeds, ensuring that all methods are compared under identical conditions. We use Stable Diffusion v1.4 [28] with a guidance scale of 7.5 and 50 inference steps, without any negative prompting. We evaluate across three domains commonly studied in prior erasure literature [2, 4, 5, 17, 20, 21, 35]; celebrities, artistic styles and objects (full details in *Suppl.*).

- **Celebrities:** We select 125 celebrities whose portraits, generated by SD v1.4 [28], are recognized by *GIPHY Celebrity Detector (GCD)* [10] with high accuracy ($> 99\%$).

It includes 25 target identities for erasure and 100 retained identities for preservation. We use GCD [10] to verify the effectiveness of identity removal.

- **Artistic Styles:** We select 125 artistic styles from the Image Synthesis Style Studies Database [21, 30], which compiles a list of artistic styles that can be replicated by SD v1.4 [28]. We select 25 target styles and 100 retained styles.
- **Objects:** Following prior erasure works [2, 4, 5, 21], we use *Imagenette* dataset [13]. It consists of 10 classes, split evenly between 5 target classes for erasure and 5 retained classes for preservation.

Experimental Protocol: To evaluate long-term behavior under repeated edits, CORE-Bench defines the **horizon** (H) as the total number of concepts sequentially erased from the model at a given evaluation point. For celebrities and artistic styles we use $H \in \{1,2,5,10,25\}$; for objects we use $H \in \{1,2,3,4,5\}$ (due to lack of availability of more object classes in *Imagenette [13]*). CORE-Bench evaluates three regimes that together cover the full spectrum of real-world concept governance scenarios.

1. **Single-Concept Erasure & Restoration.** Concepts are erased one at a time, cumulatively (e.g, at H=5, five successive single-concept erasures have been applied). This isolates the effect of each individual edit and tests stability over long sequences. Restoration is evaluated from an H=10 checkpoint by re-enabling first erased target concept, testing whether a method achieves selective recovery while keeping all remaining erasures intact.
2. **Multi-Concept Erasure & Restoration.** Concepts are erased in non-overlapping batches of five (e.g, at H=10, two such batches have been applied), measuring how simultaneous edits interfere with one another and with previously erased content. Restoration is evaluated from H=10 checkpoint by simultaneously re-enabling multiple target concepts, testing whether recovery scales without increased leakage or instability.
3. **Sequential Restore-Erase.** Starting from a checkpoint at horizon H=5, restoration and erasure steps are interleaved, one concept is restored, then new ones are erased, repeatedly simulating governance scenarios where policies evolve continuously in both directions. This is the most demanding regime and directly probes the fragility of restoration under continued editing.

Evaluation Metrics: Evaluation metrics follow recent works [2, 4, 5, 17, 20, 21] to ensure compatibility with existing literature, while extending these practices to the continual erasure and restoration setting for the first time.

For continual erasure, we adopt standard *Erasure Efficacy* ($\text{Acc}_e\downarrow$,$\text{CLIP}_e\downarrow$) and *Preservation Specificity* ($\text{Acc}_s\uparrow$,$\text{CLIP}_s\uparrow$), where detector accuracies (Acc_e,Acc_s) use GCD [10] for celebrities and ResNet-50 [11] for objects, and $\text{CLIP}_{e/s}$ are CLIP scores [12] against erased/retained concepts. Following [21], we report H_c and H_a for the measuring erasure-preservation summary.

$$H_c = \frac{2}{(1-\text{Acc}_e)^{-1} + (\text{Acc}_s)^{-1}}, \qquad H_a = \text{CLIP}_s - \text{CLIP}_e.$$

Higher values of both indicate a better erasure-preservation trade-off.

For selective restoration, we introduce three metrics that together describe a successful restoration, based on same detector accuracies:

- **Target Recovery** ($\text{Acc}_r\uparrow$): detector accuracy on the restored concepts; measures how fully the concept is recovered toward base-model behavior.
- **Resurgence rate** ($\text{Acc}_{rr}\downarrow$): detector accuracy on concepts that were previously erased and not restored now; measures unintended resurgence.

Table 1: CORE-Bench continual erasure results for celebrity and art-style domains. Higher H_c and H_a indicate better erasure-preservation trade-offs. Across both domains, **GLoCE remains consistently stable at long horizons**, while several baselines exhibit sharp degradation, especially UCE in celebrity erasure. In each row, values highlighted in **blue** are the best values at that horizon. **Closed methods** (ESD-x, UCE, and MACE) are shaded in light green, while **open methods** (SPM and GLoCE) are shaded in light blue.

Celebrity Domain (H_c)

(a) Sequential-Single

Horizon (H)	ESD-x [ICCV 23]	UCE [WACV 24]	MACE [CVPR 24]	SPM [CVPR 24]	GLoCE [CVPR 25]
1	0.78	**0.99**	0.94	0.92	0.97
2	0.60	**0.98**	0.71	0.91	**0.98**
5	0.33	0.97	0.63	0.84	**0.99**
10	0.47	0.88	0.54	0.82	**0.98**
25	0.42	0.12	0.64	0.51	**0.97**

(b) Sequential-Multiple

Horizon (H)	UCE [WACV 24]	MACE [CVPR 24]	SPM [CVPR 24]	GLoCE [CVPR 25]
5	0.97	0.64	0.84	**0.99**
10	0.89	0.58	0.82	**0.98**
15	0.88	0.46	0.80	**0.98**
20	0.79	0.48	0.66	**0.97**
25	0.11	0.45	0.51	**0.97**

Art-Style Domain (H_a)

(c) Sequential-Single

Horizon (H)	ESD-x [ICCV 23]	UCE [WACV 24]	MACE [CVPR 24]	SPM [CVPR 24]	GLoCE [CVPR 25]
1	1.75	1.11	5.37	**7.57**	6.17
2	0.65	1.84	3.05	3.43	**4.00**
5	0.60	0.74	-0.13	**5.11**	4.02
10	0.98	-0.17	0.27	**6.44**	3.34
25	-0.46	-0.82	0.19	-0.31	**2.52**

(d) Sequential-Multiple

Horizon (H)	UCE [WACV 24]	MACE [CVPR 24]	SPM [CVPR 24]	GLoCE [CVPR 25]
5	0.70	4.61	**5.11**	4.02
10	-0.21	2.08	**6.44**	3.34
15	-0.37	2.23	**5.92**	2.80
20	-0.11	1.67	**4.74**	2.62
25	-1.29	1.55	-0.31	**2.52**

- **Retain Stability** $(\text{Acc}_{ret}\uparrow)$: detector accuracy on never-erased retained concepts; measures collateral drift caused by the restoration update.

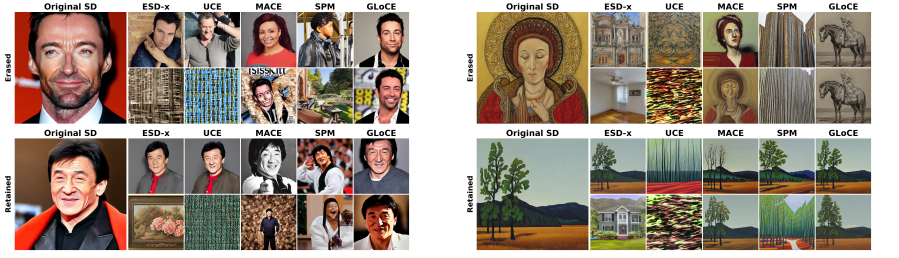
6 Experimental Evaluation

We evaluate continual concept erasure and selective restoration using CORE-Bench across three domains (celebrities, artistic styles, objects). For continual erasure, we compare ESD-x [4], UCE [5], MACE [21], SPM [22], and GLoCE [17] under both Sequential-Single and Sequential-Multiple regimes. For restoration, we evaluate closed-form editing methods, UCE & SPEED [20] and training-based method, ESD-x, as representative methods from each design class.

6.1 Continual Erasure

Table 1 and Figure 2 report continual erasure results for the celebrity and artistic-style domains. Object-domain results follow the same trend and are reported in the *Suppl.* due to space constraints. Three consistent findings emerge across domains and horizons.

Gating-based methods are the most stable. As expected, SPM [22] and GLoCE [17] remain stable at all horizons in both regimes. Their modular adapter design prevents inter-edit interference by construction. Each new concept receives its own adapter without modifying



(a) Celebrities: erased *Hugh Jackman*, retained *Jackie Chan*.

(b) Artists: erased *Carlo Crivelli*, retained *A.J. Casson*.

Figure 2: Qualitative comparison of generations from the base model and different concept-erasure methods for celebrities and artistic styles (Sec. 6.1). Each case includes one erased concept and one retained concept; for each concept, the first row corresponds to $H = 1$ and the second row corresponds to $H = 25$. Adapter based methods, GLoCE and SPM, are performing better than other methods at $H = 25$.

previously learned ones. GLoCE is particularly strong on celebrities, consistent with its localized inference-time gating. On artistic styles, GLoCE degrades more than on celebrities, which we attribute to artistic styles being more globally distributed in representation space and less amenable to localized gating.

Closed-form methods degrade at long horizons. UCE [6] is competitive up to H=10 but collapses at H=25 for celebrities, consistent with successive closed-form updates accumulating interference in the cross-attention weight space without an explicit mechanism to protect prior erasures. MACE [21] is more stable than UCE at long horizons, likely due to its per-concept LoRA fusion step that explicitly protects preserved directions.

Training-based methods are weakest overall. ESD-x [4] degrades rapidly with horizon due to cumulative parameter drift from successive fine-tuning steps [13]. CCRT [9], designed for short sequences, drops steeply beyond its intended range (H_c=0.81 at H=1 to 0.14 at H=5 for celebrities, H_c=0.78 at H=1 to 0.28 at H=5 for objects), confirming that its calibration term is insufficient at even moderately longer horizons.

6.2 Selective Restoration Results

We evaluate selective restoration for UCE [6] and SPEED [20] using Eqs. (8) and (9), respectively, from a horizon-H=10 celebrity checkpoint, restoring either a single identity (*Hugh Jackman*) or three identities (*Hugh Jackman*, *Hillary Clinton*, *Stan Lee*). Table 2(a,b) reports accuracy on restored identities, other erased (non-restored) identities at the same horizon, and a held-out retained set. Both methods consistently recover the target identities toward base-model accuracy while leaving other erased and retained performance largely unchanged. Table 2(a) shows the results of single-concept restoration of *Hugh Jackman* from H=10 checkpoint using Eq. (7) of ESD-x. Multi-concept restoration is not evaluated for ESD-x because the method natively supports only single-concept erasure.

Concept resurgence in noise-prediction methods: A striking difference emerges between method classes. ESD-x exhibits severe leakage upon restoration; Acc_{rr} jumps from 0.333 to

Table 2: Comparison of ESD-x (ICCV 2023), UCE (WACV 2024), SPEED (ICLR 2026) on the task of Selective restoration(Sec. 6.2). Higher Acc_r, Acc_{ret} are better, while lower Acc_{rr} indicates less leakage. HJ = Hugh Jackman, HC = Hillary Clinton, and SL = Stan Lee. Object restoration is evaluated at H=5 by restoring *cassette player* and *church*. Values in orange show the presence of concept resurgence.

Celebrity Domain

(a) Single-Concept Restoration: HJ

Stage / Method	$\text{Acc}_r \uparrow$	$\text{Acc}_{rr} \downarrow$	$\text{Acc}_{ret} \uparrow$
Base	0.925	1.000	0.980
ESD-x erased	0.025	**0.333**	0.210
ESD-x restored	0.850	**0.867**	0.920
UCE erased	0.025	0.067	0.770
UCE restored	0.775	0.067	0.810
SPEED erased	0.000	0.072	0.831
SPEED restored	1.000	0.074	0.854

(b) Multi-Concept Restoration: HJ, HC, SL

Stage / Method	$\text{Acc}_r \uparrow$	$\text{Acc}_{rr} \downarrow$	$\text{Acc}_{ret} \uparrow$
Base	0.975	1.000	0.980
UCE erased	0.008	0.086	0.770
UCE restored	0.908	0.086	0.790
SPEED erased	0.000	0.093	0.831
SPEED restored	1.000	0.131	0.823

Object Domain

(c) Multi-Concept Restoration: Cassette Player, Church

Stage / Method	$\text{Acc}_r \uparrow$	$\text{Acc}_{rr} \downarrow$	$\text{Acc}_{ret} \uparrow$
Base	0.800	0.822	0.880
UCE erased	0.000	0.000	0.720
UCE restored	0.833	0.022	0.733
SPEED erased	0.180	0.027	0.848
SPEED restored	0.785	0.013	0.832

0.867 (highlighted in orange in Tab.2(a)) after restoring a single identity, meaning that the restoration update unintentionally reactivates nearly all other erased concepts. By contrast, UCE and SPEED maintain consistently low leakage ($\text{Acc}_{rr} \approx 0.067$ & ≈ 0.074, respectively), demonstrating that the restoration is genuinely selective. This disparity aligns with prior observations of *concept resurgence* [6, 7, 31], where fine-tuning updates interfere with earlier edits and unintentionally reactivate suppressed concepts. We also evaluate restoration in the object domain using the *Imagenette* [13] object split. Starting from an object-erased checkpoint at H=5, where all five target objects are erased, we restore the first two target objects; *cassette player* and *church*. The remaining erased objects, *French horn*, *garbage truck*, and *parachute*, are used to measure leakage, while *tench*, *gas pump*, *golf ball*, *English springer*, and *chain saw* are used as retained objects. We report object-domain multi-concept restoration results in Table 2(c).

Concept-conditioned response analysis: To further examine the selectivity of closed-form edits, we analyze how restoration changes the cross-attention responses associated with different concept embeddings for UCE. For each concept c and each UCE-edited cross-attention layer l, we compute the normalized response movement

$$M_l(c) = \frac{\|f_l^{\text{restored}}(c) - f_l^{\text{erased}}(c)\|_2}{\|f_l^{\text{erased}}(c) - f_l^{\text{base}}(c)\|_2 + \varepsilon}, \tag{10}$$

where $f_l(c)$ denotes the output of layer l when the text embedding of concept c is passed through the corresponding UCE-edited projection. This quantity measures how much the

(a) UCE Identity restoration example. (b) UCE Object restoration example.

Figure 3: Selective restoration results (Sec. 6.2). Top: Original, Middle: Erased, Bottom: Restored. In (a), *Hillary Clinton* and *Stan Lee* are restored; *Peter Dinklage* remains erased; *Aaron Paul* is retained throughout. In (b). *Cassette Player* and *Church* are restored; *Parachute* remains erased; *Gas Pump* is retained throughout. Restored concepts recover base-model fidelity while non-restored erased concepts remain suppressed and retained concepts are unaffected.

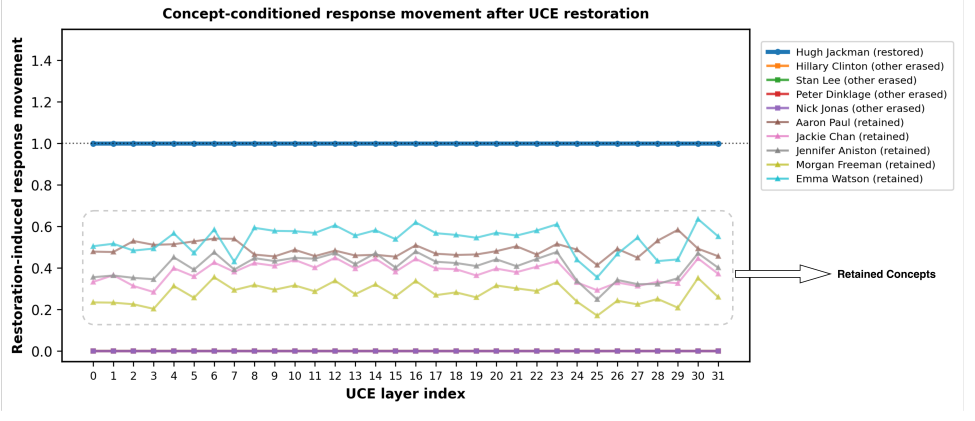


Figure 4: Concept-conditioned response movement after UCE restoration. Restoration moves the target concept back toward the base-model response, while non-restored erased concepts remain close to the erased checkpoint. Retained concepts exhibit only moderate movement, showing that UCE restoration acts selectively.

restoration update moves a concept response away from the erased checkpoint, normalized by the amount of change originally introduced by erasure. A larger value indicates that restoration substantially changes the concept response, while a smaller value indicates that the concept remains close to the erased model. Fig. 4 plots the mean normalized response movement across three groups: the restored concept, the non-restored erased concepts, and the retained concepts. The restored concept shows consistently higher response movement across UCE-edited layers, while the non-restored erased concepts remain close to the erased checkpoint. This supports the interpretation that UCE restoration does not globally roll back the erased model, but instead selectively modifies the restored concept direction. Retained concepts show moderate movement, but their detector accuracy and qualitative generations remain stable, suggesting that this response-level change does not lead to substantial se-

Table 3: Sequential restore–erase results for UCE and SPEED. HJ = Hugh Jackman, HC = Hillary Clinton, SL = Stan Lee. CP = cassette player, CH = church, and FH = French horn. Higher Acc_r and Acc_{ret} are better, while lower Acc_{rr} indicates less leakage. High final restored accuracies (in **blue**) across domains and methods indicate that closed-form restoration embeds the recovered concept strongly enough to survive further edits.

(a) Celebrity sequential restore-erase

Stage	**UCE** [WACV 24]					**SPEED** [ICLR 26]				
	Restored identities			**Others**		**Restored identities**			**Others**	
	HJ ↑	**HC** ↑	**SL** ↑	Acc_{rr} ↓	Acc_{ret} ↑	**HJ** ↑	**HC** ↑	**SL** ↑	Acc_{rr} ↓	Acc_{ret} ↑
Base	1.000	1.000	0.750	0.969	0.950	1.000	1.000	0.750	0.969	0.950
Erased ($E_{H=5}$)	0.000	0.000	0.000	0.104	0.800	0.000	0.000	0.000	0.000	0.816
Restore HJ	1.000	0.000	0.000	0.104	0.750	1.000	0.000	0.000	0.000	0.840
Erase +3 after HJ	0.250	0.000	0.000	0.083	0.800	1.000	0.000	0.000	0.000	0.795
Restore HC	0.500	1.000	0.000	0.167	0.775	1.000	1.000	0.000	0.000	0.795
Erase +3 after HC	0.500	1.000	0.000	0.125	0.800	0.960	1.000	0.000	0.009	0.795
Restore SL	**0.500**	**1.000**	**1.000**	0.125	0.800	**0.960**	**1.000**	**1.000**	0.015	0.795

(b) Object sequential restore-erase

Stage	**UCE** [WACV 24]					**SPEED** [ICLR 26]				
	Restored objects			**Others**		**Restored objects**			**Others**	
	CP ↑	**CH** ↑	**FH** ↑	Acc_{rr} ↓	Acc_{ret} ↑	**CP** ↑	**CH** ↑	**FH** ↑	Acc_{rr} ↓	Acc_{ret} ↑
Base	1.000	0.800	1.000	0.733	0.880	1.000	0.800	1.000	0.733	0.880
Erased ($E_{H=3}$)	0.000	0.000	0.000	0.000	0.773	0.000	0.000	0.040	0.000	0.920
Restore CP	0.815	0.000	0.000	0.000	0.720	0.800	0.080	0.040	0.105	0.893
Erase garbage truck	0.786	0.000	0.000	0.000	0.747	0.820	0.160	0.040	0.080	0.910
Restore CH	0.804	0.867	0.000	0.033	0.787	0.820	0.760	0.000	0.040	0.930
Erase parachute	0.795	0.833	0.000	0.000	0.793	0.860	0.680	0.000	0.000	0.856
Restore FH	**0.800**	**0.867**	**1.000**	0.000	0.785	**0.800**	**0.760**	**1.000**	0.000	0.840

mantic drift. Fig. 3 provides qualitative examples supporting the same behavior; restored concepts reappear, non-restored erased concepts remain suppressed, and retained concepts are preserved.

6.3 Sequential Restore-Erase Setting Results

Prior restoration evaluation in Sec. 6.2 fixed the edit sequence and restore from a static checkpoint. In sequential restore-erase setting, restoration and erasure steps are *interleaved*. Every new erasure risks de-stabilizing concepts that were already restored and every restoration risks leaking concepts that were erased after the starting checkpoint. To our knowledge, no prior work has evaluated concept restoration under this extremely challenging regime. Since this experiment requires the simultaneous erasure of multiple concepts, ESD-x was omitted from the evaluation due to its inability to support multi-concept erasure.

For celebrities, we start from the checkpoint at H=5, where the erased identities are *Hugh Jackman*, *Hillary Clinton*, *Stan Lee*, *Anne Hathaway*, and *Meryl Streep*. We then follow three restore–erase cycles: restore *Hugh Jackman*, erase *Jennifer Lopez*, *Adriana Lima*, and *Nick Jonas*; restore *Hillary Clinton*, erase *Bob Marley*, *Peter Dinklage*, and *Octavia Spencer*; finally, restore *Stan Lee*. For objects, we start from the checkpoint at H=3, where the erased objects are *cassette player*, *church*, and *French horn*. We perform the same interleaved pattern; restore *cassette player*, erase *garbage truck*; restore *church*, erase *parachute*; finally, restore *French horn*. Table 3 and Fig. 5 summarize the restore-erase results.

(a) UCE Celebrity restore-erase cycles. (b) SPEED Object restore-erase cycles.

Figure 5: Sequential restore-erase cycles at various stages (Sec. 6.3). In (a), rows show: (1) *Hillary Clinton* restored, (2) *Stan Lee* restored, (3) *Nick Jonas* erased, (4) *Peter Dinklage* erased, and (5) *Aaron Paul* retained throughout. In (b), rows show: (1) *cassette player* restored, (2) *church* restored, (3) *French horn* restored, (4) *garbage truck* erased, and (5) *gas pump* retained throughout. Restored concepts remain stable across subsequent erase steps, and newly erased concepts are suppressed without reactivating prior restorations.

Two findings emerge from these results.

(i) *Restored concepts remain stable under subsequent erasure.* Once a concept is restored, further erase steps do not suppress it back to zero. It shows that closed-form restoration embeds the recovered concept durably enough to survive further edits, rather than merely producing a fragile, transient recovery.

(ii) *Retained concepts are unaffected.* Acc_{ret} remains stable across all stages for both methods and both domains, confirming that neither restoration nor the subsequent erasures produce collateral drift on never-targeted concepts.

7 Conclusion and Takeaways

We introduced *CORE-Bench*, the first standardized benchmark for evaluating continual concept erasure and restoration in text-to-image diffusion models across three domains and edit horizons up to $H{=}25$. Unlike prior static benchmarks, CORE-Bench jointly tracks erasure persistence, semantic drift on retained concepts, and selective restoration under long-horizon governance scenarios.

Benchmarking existing erasure methods reveals key trade-offs: PEFT-based gating methods (SPM, GLoCE) remain structurally stable but are unsuitable for open-weight deployment; closed-form editors (UCE, SPEED) perform well up to $H{=}10$ before accumulating interference; and training-based methods degrade most rapidly. No current method satisfies open-weight deployment, long-horizon stability, and selective restoration simultaneously.

Additionally, we formally define and evaluate selective concept restoration; re-enabling an erased concept without compromising other removals. We propose a constrained counter-edit formulation for both closed-form and training-based erasers, alongside three dedicated evaluation metrics. Results show closed-form editors achieve near-zero leakage, whereas training-based methods suffer severe concept resurgence. CORE-Bench provides a unified foundation for building governable generative AI systems.

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